



BIM Portfolio

Where Virtual Models Shape Real-World Solutions — My Journey in BIM.

Abinayasri Samadharman

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About Me

Creative and detail-oriented BIM Modeler. Skilled in Revit and digital coordination workflows.

2

What I Do

Develop precise 3D models across disciplines. Ensure accurate clash-free coordination.

3

My Experience

Hands-on experience in commercial, residential, and infrastructure projects.

4

My work

Includes Revit modelling, Dynamo scripts, and Navisworks coordination outputs.

5

Exhibition

Highlighted projects with smart detailing and clean coordination deliverables.

6

Next Projects

Exploring Dynamo automation and Power BI analytics for BIM advancement.

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I'm a Civil Engineer turned BIM professional who blends precision with creativity to craft intelligent, data-driven models. My journey across metro rail and residential projects has strengthened my skills in coordination, problem-solving, and multidisciplinary collaboration.

I see BIM as more than modeling — it's a storytelling process that connects design, engineering, and construction into one cohesive system.

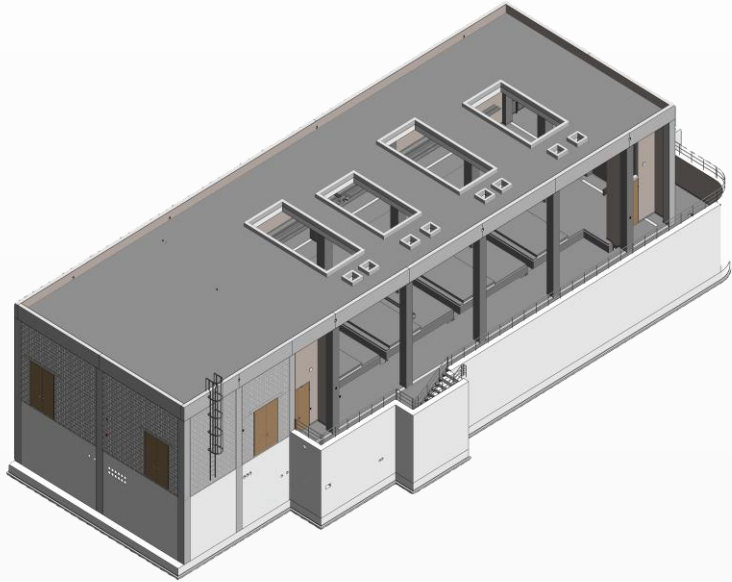
I'm passionate about using technology to bring clarity to complexity and redefine how we design and build.

About Me



Karur, Tamil Nadu, India.

2



As a BIM Modeler developed detailed 3D BIM models (LOD 300–500) for architecture and structure disciplines.

- ❖ Perform clash detection and coordination using Navisworks.
- ❖ Automate workflows and data modelling with Dynamo scripts.
- ❖ Extract quantities, drawings, and schedules from BIM models.
- ❖ Ensure accuracy and collaboration across project stakeholders.

What I Do

My Experience

APR-2024 → PRESENT	KALEIDOSCOPE Handled urban and commercial BIM projects with clash detection and model coordination.
MAY-2023 → APR-2024	PINNACLE TECH SERVICES (PTS) Modelled Netherlands residential buildings and converted 2D CAD to Revit BIM models

My Education

2019 2023	GCE COLLEGE Completed B.E. Civil with a structural design focus.
2022 2023	NOVATR Certified in Revit, Navisworks & Dynamo essentials.

My Skills

 Revit Arch/Str	■ ■ ■ ■ ○
 AutoCAD 2D	■ ■ ■ ■ ○
 Navisworks	■ ■ ■ ■ ○
 Dynamo	■ ■ ■ ○ ○
 Plugin(Di-roots)	■ ■ ■ ■ ○
Synchro 4D	■ ■ ■ ○ ○

Precisions

Delivering highly detailed and accurate BIM models that meet LOD standards across all stages.

Coordination

Ensuring clash-free, multi-disciplinary collaboration for efficient project delivery.

Documentation

Creating clean, standardized drawings and schedules directly from the model.

Automation

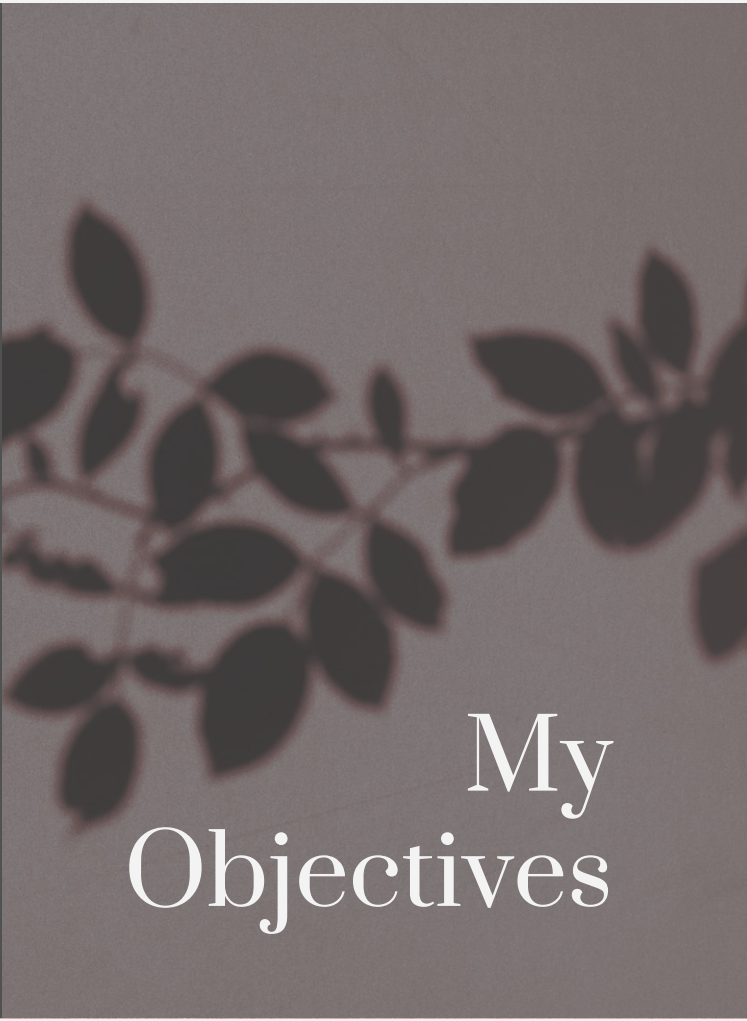
Utilizing Dynamo and BIM tools to optimize repetitive tasks and workflows.

Growth

Continuously upgrading skills to contribute to smarter and more digital construction.

Adaptability

Quickly adjusting to evolving project requirements, tools, and client standards in BIM workflows.



My Objectives



20

Handled basic site layout modelling, topography, and site components for BIM setup.



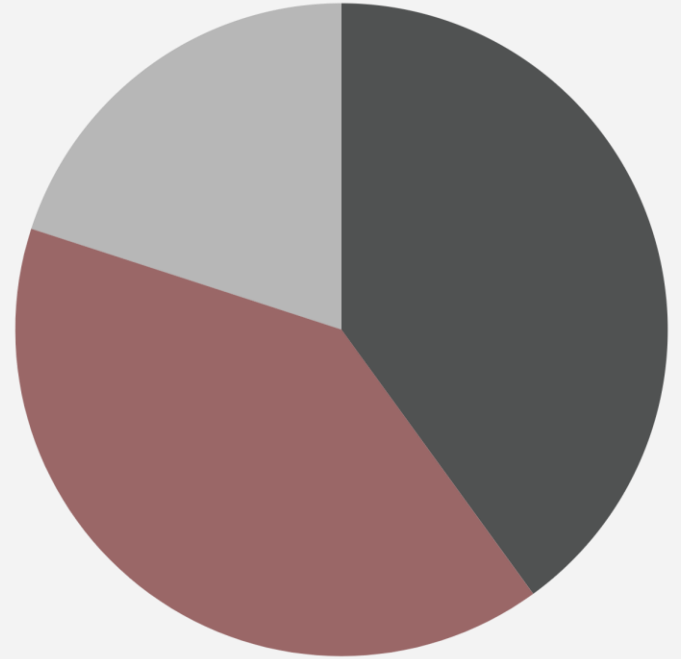
40

Created construction sheets, annotations, schedules, and legends based on Revit models.



40

Developed architecture/structure/MEP models and custom Revit families for projects.



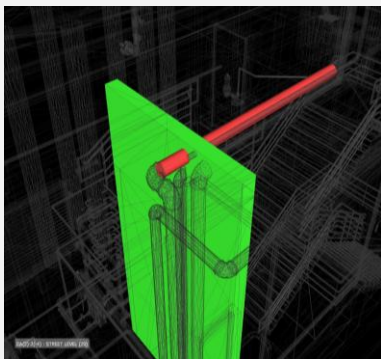
Percentages
of Effort Across BIM Tasks



My Work

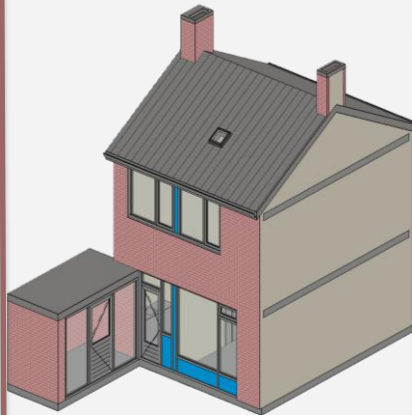
Includes Revit modelling, Dynamo scripts, and Navisworks coordination outputs.

4



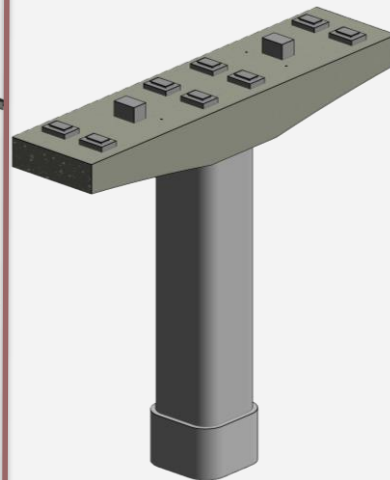
Clash Coordination

Executed clash detection using Navisworks for MEP-Structure models, ensuring conflict-free construction.



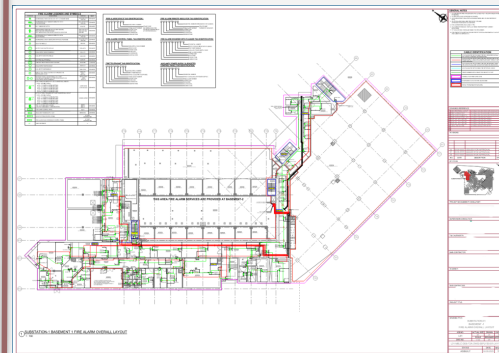
Residential Modelling

Developed LOD 300 Revit models for villas and apartment buildings with structural and architectural detailing.



Family Creation

Designed custom Revit families for fixtures, furniture, and equipment with parametric controls.



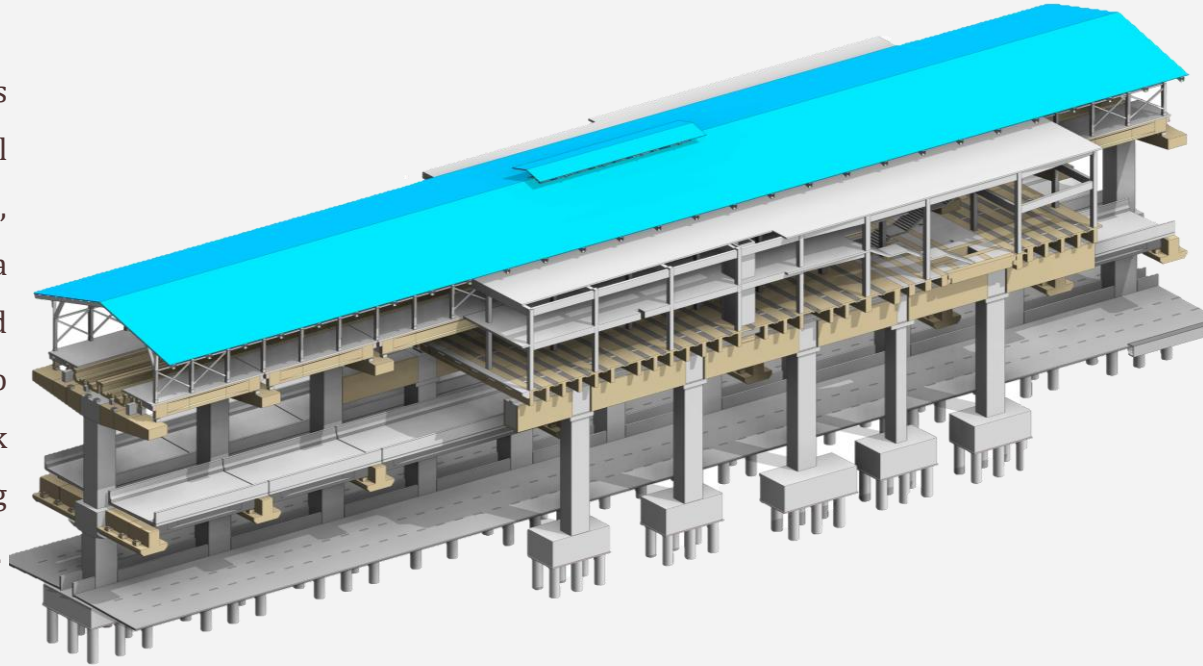
As-Built Documentation

Prepared as-built BIM models by updating site markups and scanned data into Revit models.

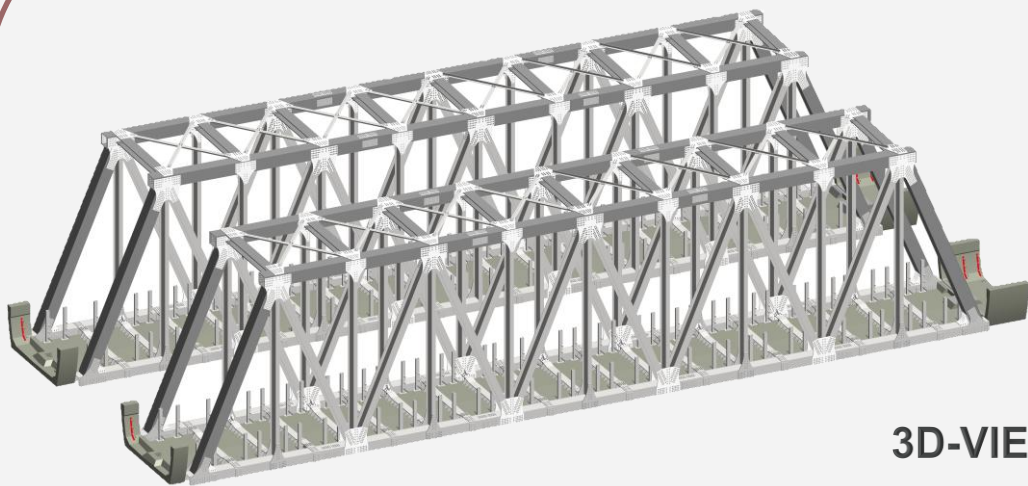
My Work

Designing with Precision, Modeling with Purpose

BIM (Building Information Modelling) is more than just 3D modelling—it's a digital process that shapes the way architecture, engineering, and construction come to life. As a BIM Modeller, I integrate data, design, and collaboration to transform conceptual ideas into functional, constructible realities. My work ensures projects are not only visually compelling but also technically sound and sustainably driven.



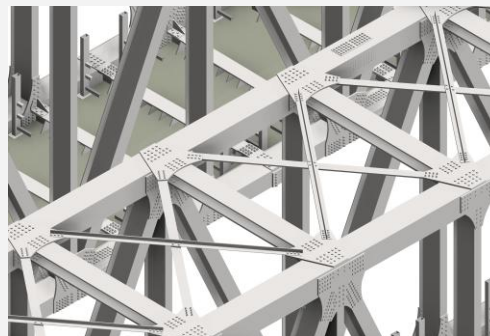
"Bridging Ideas with Intelligent Design"



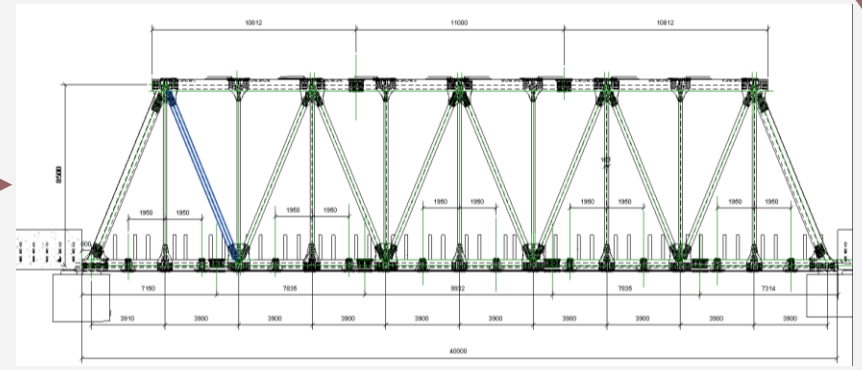
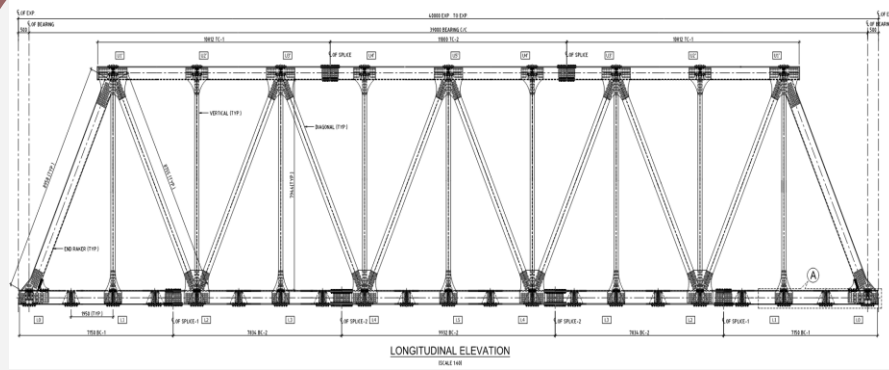
3D-VIEW



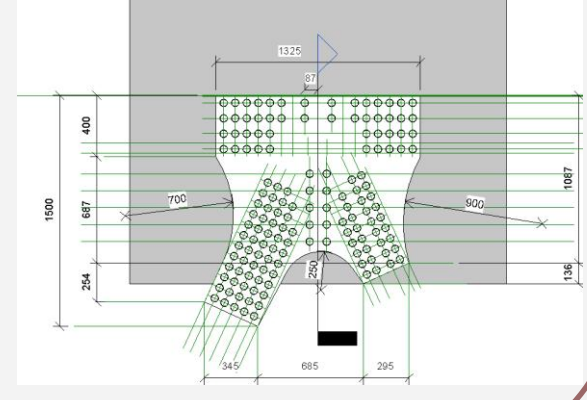
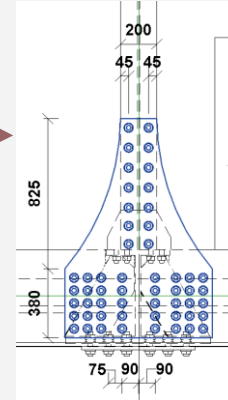
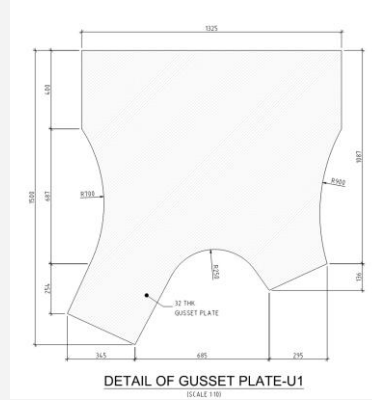
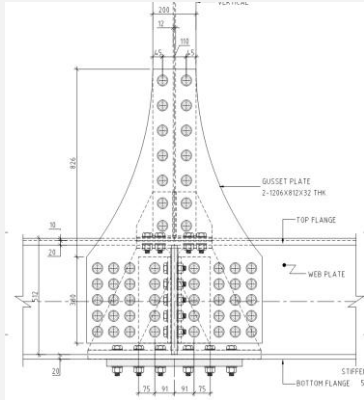
AS-BUILT

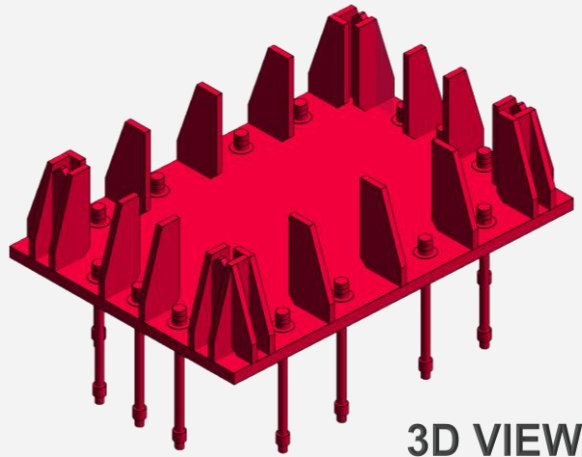
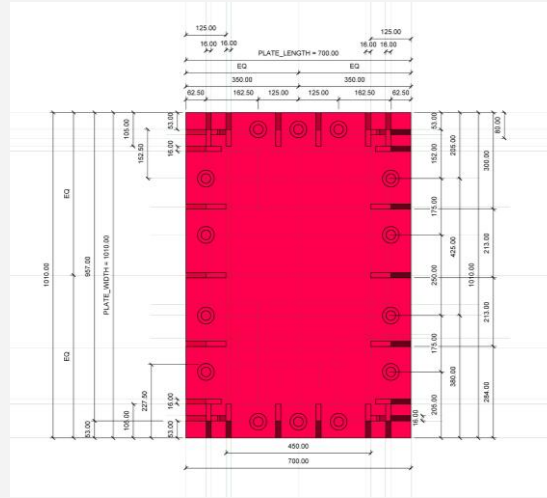
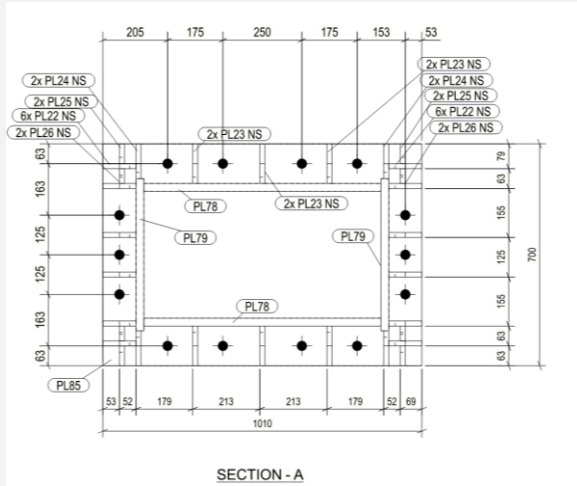


Modeled a 40m curved steel span with complex connections, developing custom workflows in Revit to overcome geometry limitations. Delivered a precise as-built model that met client expectations with high accuracy and quality.



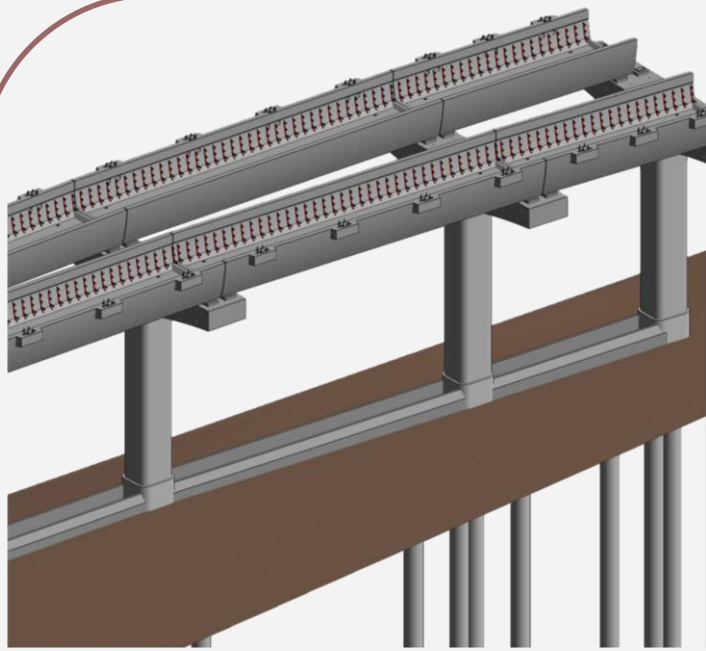
INPUT → MODEL OUTCOME



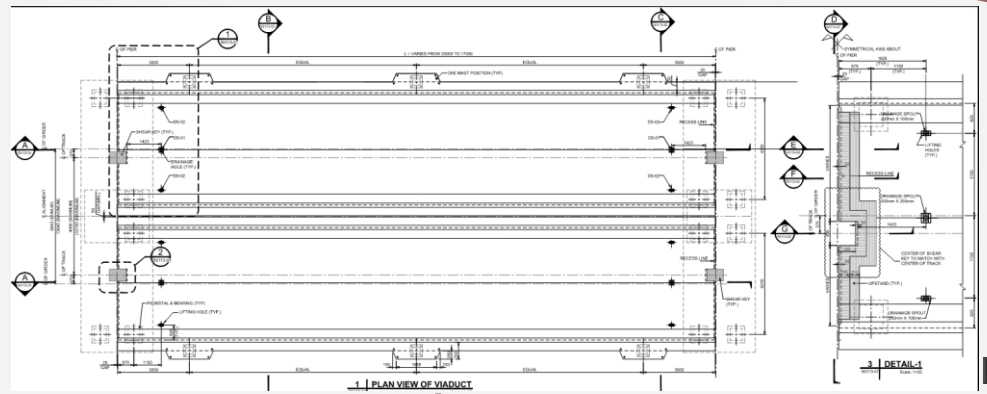
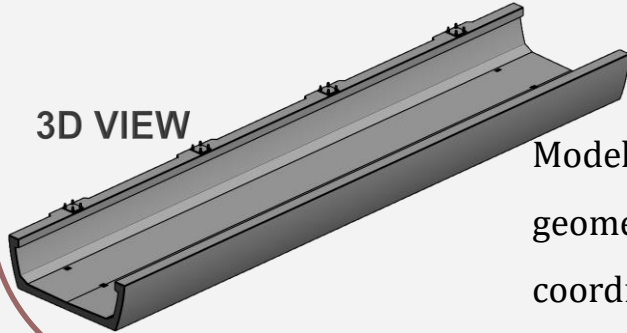


Developing parametric families may be complex at the start, but it streamlines the process for creating similar families later.

Before creating a new family, I always check if an existing parametric family can be adapted, ensuring efficiency, accuracy, and time savings.

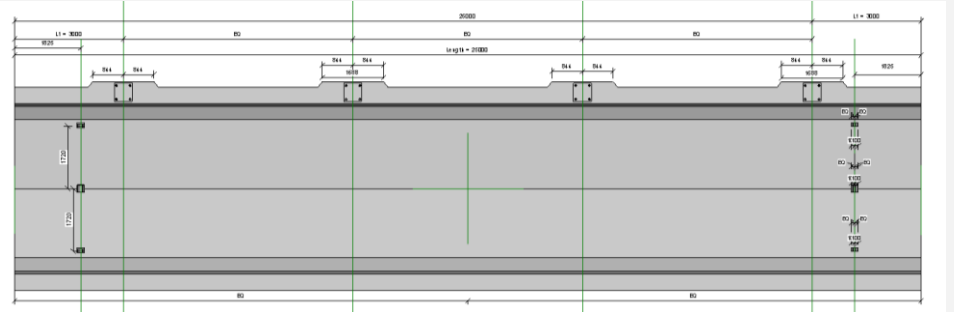


3D VIEW

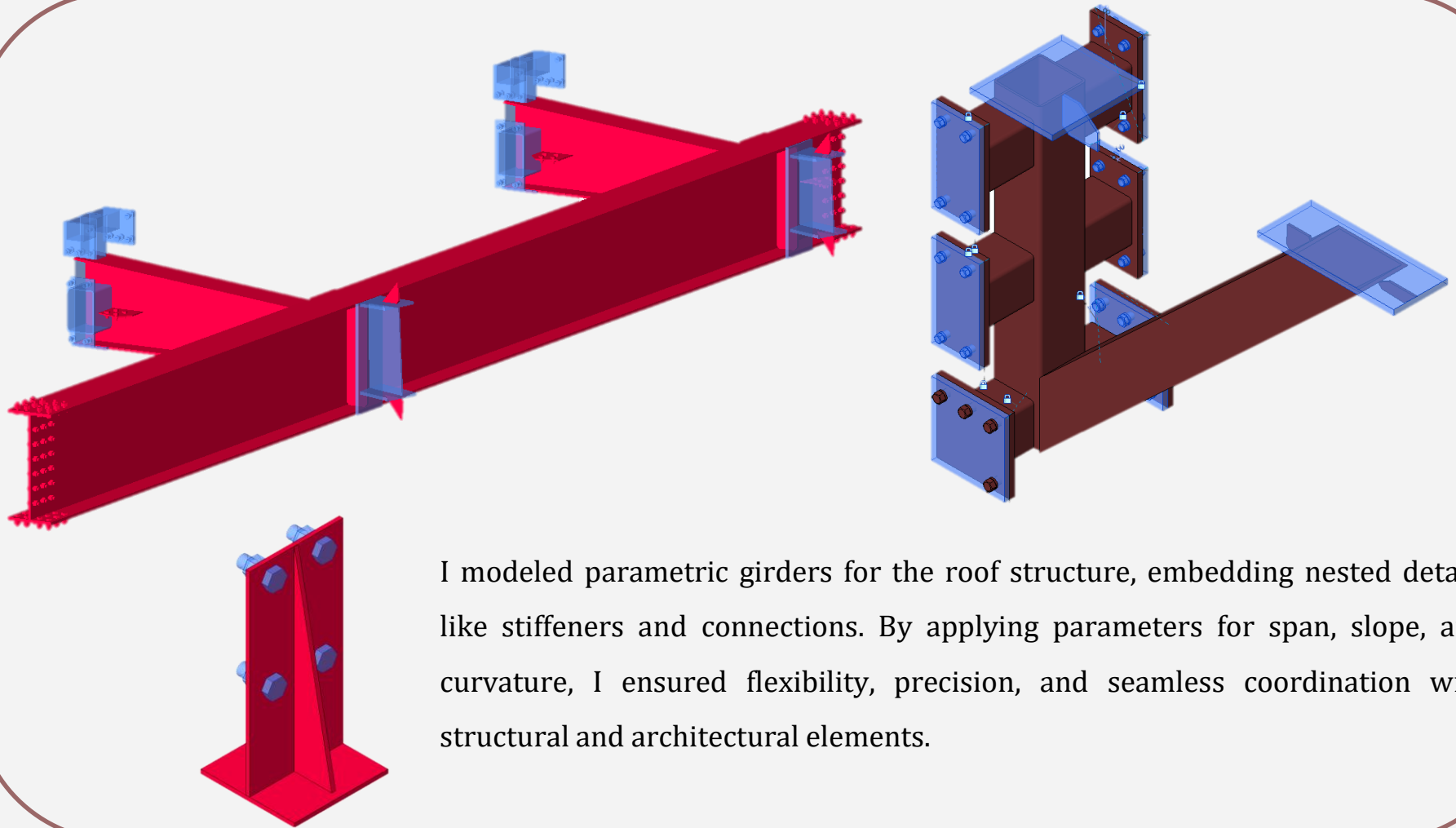


INPUT

OUTCOME

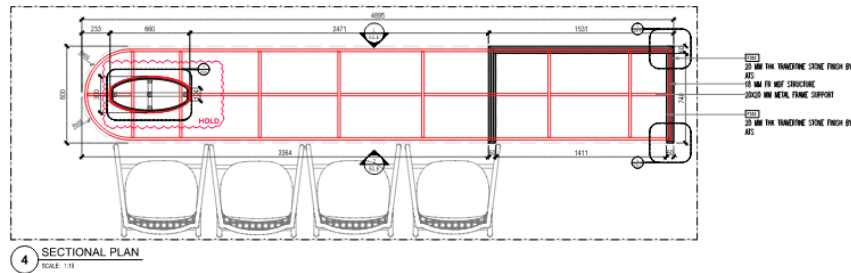


Modeled and delivered the viaduct track system with accurate alignment geometry, plinths, and fastening assemblies. Developed parametric components, coordinated with structural/MEP disciplines, and produced LOD 300-400 models with drawings and as-built documentation.

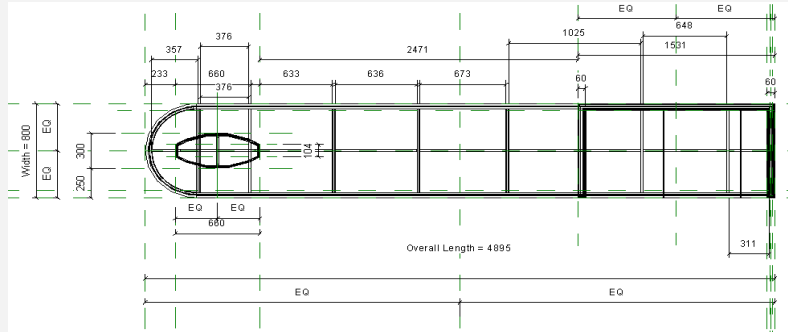


I modeled parametric girders for the roof structure, embedding nested details like stiffeners and connections. By applying parameters for span, slope, and curvature, I ensured flexibility, precision, and seamless coordination with structural and architectural elements.

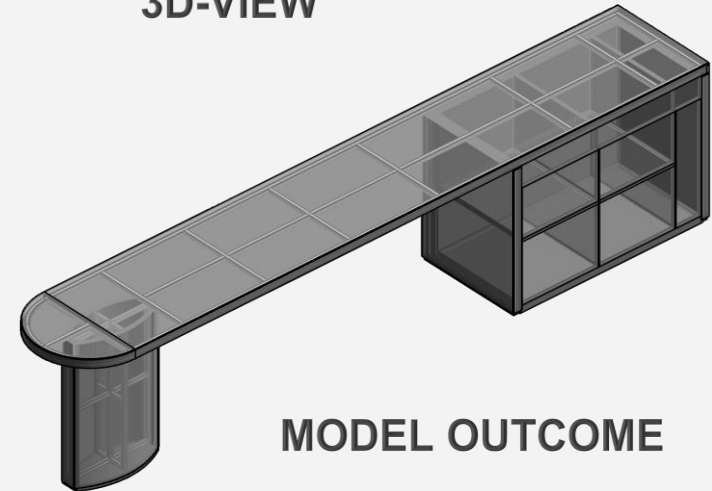
INPUT



3D-VIEW

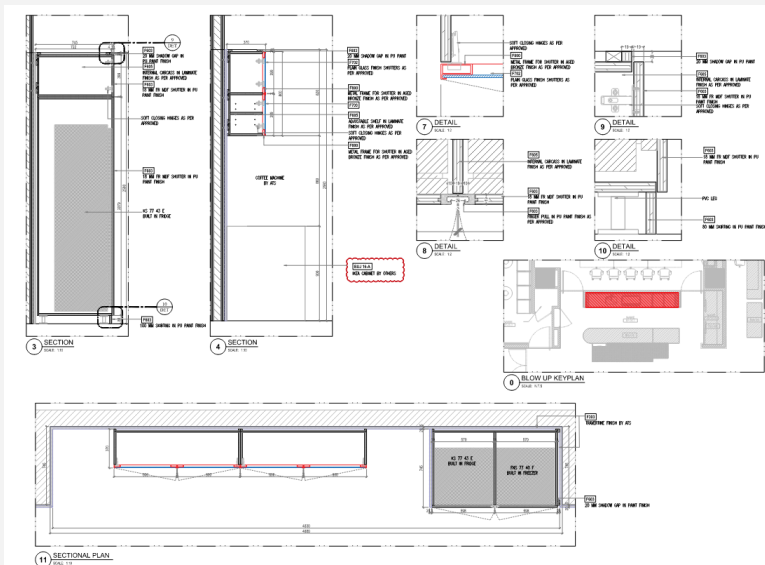


REF VIEW

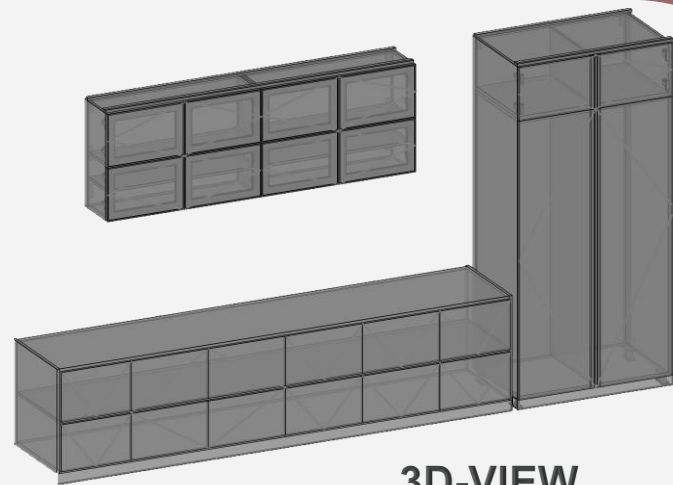


MODEL OUTCOME

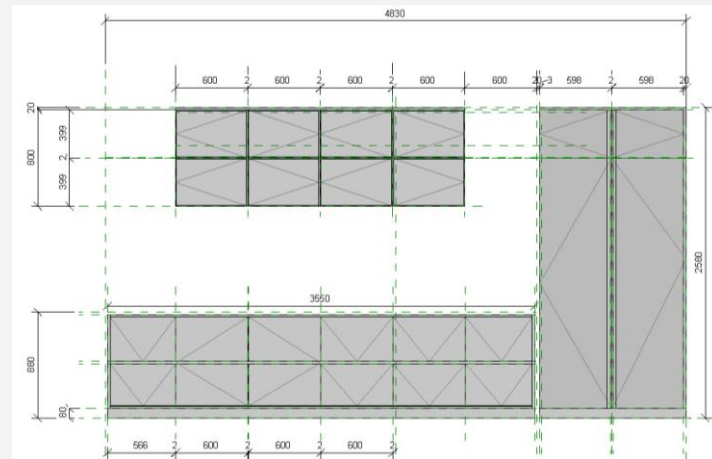
INPUT



AS-BUILT IMAGE

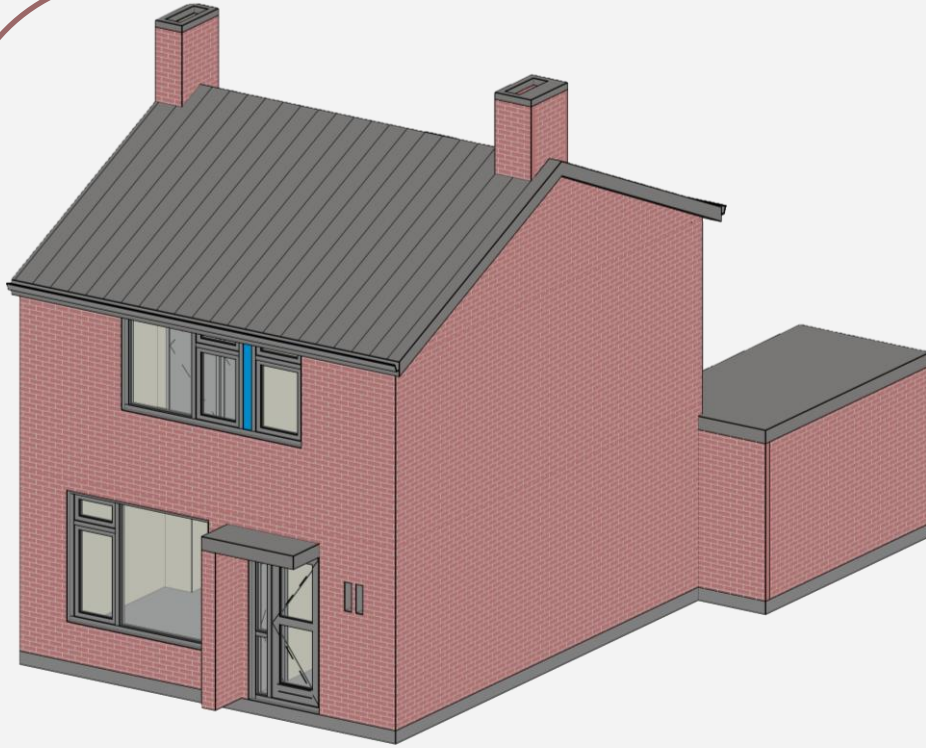


3D-VIEW



FRONT VIEW

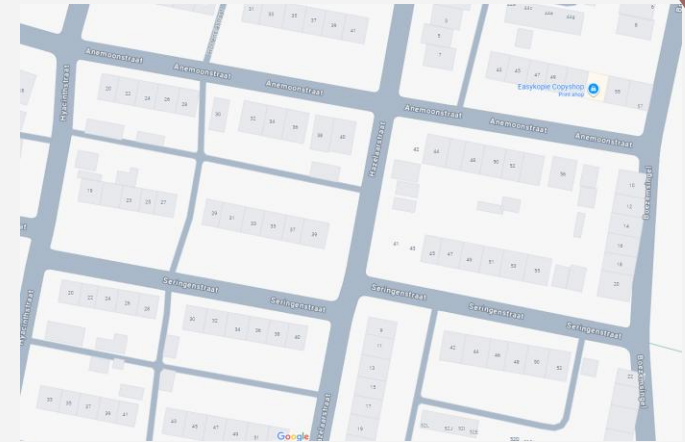
Google - to LOD 500 updating for documentation



Delivered a government-mandated residential project in the Netherlands, maintaining Revit models at LOD 300–500. Ensured all design updates and site changes were reflected accurately, serving as the single source of truth for government documentation, approvals, and facility management.

Gebouw Schedule work		
Material mentioned in some schedule are Default, if specific given in sourcethen need to assign accordingly		
While placing Radiator, black part should be facing wall		
Schedule name	Parameter	Action
IfcAirToAirHeatRecovery_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcAlarm_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcAudioVisualAppliance_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcBoiler_Gebouw ILS	Collectief	Just check if all fields in schedule are filled, only applicable when components are placed Tick must be On
IfcBuildingElementProxy_Gebouw ILS		Nothing to do
IfcBurner_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcColumn_Gebouw ILS	All parameters as it is, only Breedte/Diameter to be checked	
	Breedte	Fill 0 if column is circular
	Diameter	Fill 0 if column is not circular
IfcCurtainWall_Gebouw ILS	Exterior Tick	exterior tick on while for exterior columns and off for Interior Columns
	Exterior Tick	exterior tick on while for exterior Door/Windows and off for Interior Door/Windows
	Exterior tick	exterior tick on while for exterior Door and off for Interior Door
IfcDoor_Gebouw ILS	Gemeenschappelijk	on While Gemeenschappelijk for Common area Door
	Soort deur	Fill According to finction, Check "RH_Stepwise" document page 1 point 12
	Toegangsdeur	on While Toegangsdeur for only Entrée Door
IfcElectricDistributionBoard_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcFireSuppressionTerminal_Gebouw ILS		Just check if all fields in schedule are filled, only applicable when components are placed
IfcFurniture (Privacy screen)_Woonstad		Just check if all fields in schedule are filled, only applicable when components are placed
ILSIfcFurniture_Gebouw ILS	IfcExportAs	IfcFurniture
	Assembly Code	It should be filled correctly
IfcPump_Gebouw ILS	ObjectType	It should be filled correctly
		Nothing to do (Keep as default)
IfcRailing_Gebouw ILS	Afsluitbaar	Off
	Exterior Tick	On/Off as per int/ext railing
IfcRoof_Gebouw ILS	Oppervlakte	Filled by Dynamo (Railing area and length)
	Daktype	Need to fill according to type of roof
IfcSanitaryTerminal_Gebouw ILS	ObjectType	Filled by Dynamo (ObjectType)
IfcSensor_Gebouw ILS		Nothing to do (Keep as default)
IfcSlab_Gebouw ILS	Exterior Tick	exterior tick on while for exterior floor and off for Interior floor

The street view of building

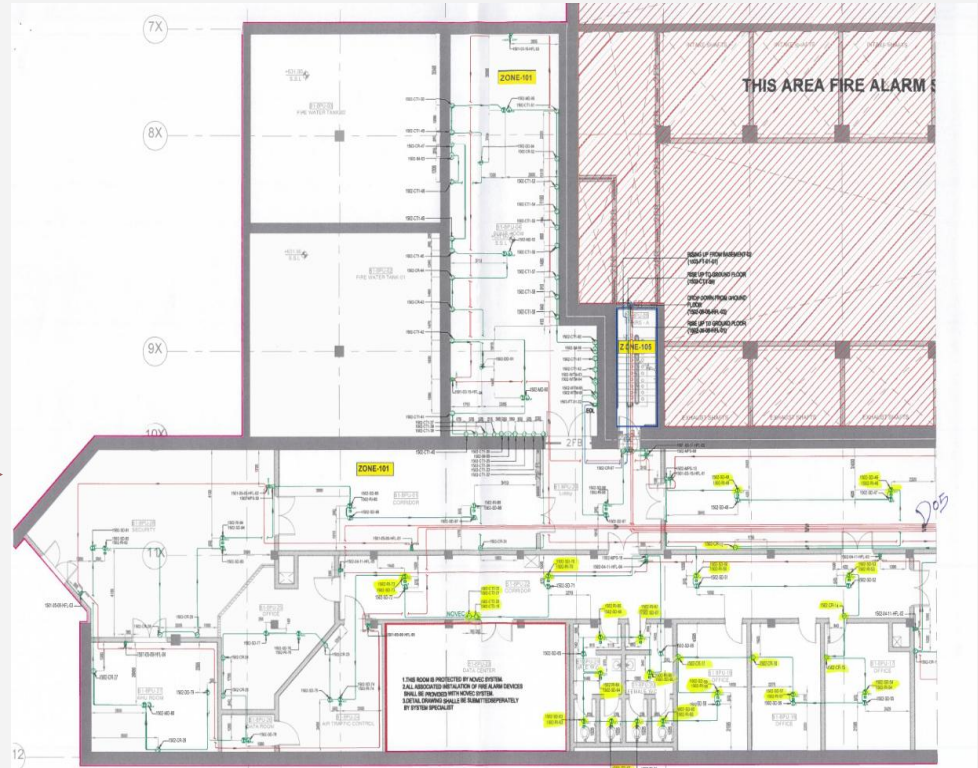


Ensured client requirements by updating Revit schedules (LOD 300–500) with accurate IFC parameters, correct material assignments, and Dynamo automation for repetitive tasks such as railing length and area.

Old Complex number	Complexnumber	Address (Straat)	House Number (Huisnummer)	House letter (Toevoeging)	House number addition (Volgnummer)	Zip code (Postcode)	City (Plaats)	VHE number	Bag residence id (BAG objectid)	Bag building id (BAG pand)	Type	Building Type
31	103108	Hazelaarstraat	9			3261 XC	OUD-BEIJERLAND	103108001				
31	103108	Hazelaarstraat	11			3261 XC	OUD-BEIJERLAND	103108002				
31	103108	Hazelaarstraat	13			3261 XC	OUD-BEIJERLAND	103108003				
31	103108	Hazelaarstraat	15			3261 XC	OUD-BEIJERLAND	103108004				
31	103108	Hazelaarstraat	17			3261 XC	OUD-BEIJERLAND	103108005				
31	103108	Hazelaarstraat	19			3261 XC	OUD-BEIJERLAND	103108006				
31	103108	Hazelaarstraat	21			3261 XC	OUD-BEIJERLAND	103108007				
31	103108	Hazelaarstraat	23			3261 XC	OUD-BEIJERLAND	103108008				

Details of residential building

Work Inspection Request			
Request Ref.:	L01-MBLC-CON-CIV-WIR-000067	Revision:	R02
Project:	L01	Submission Date:	07/11/20
Client:	CPPA	Inspection Date:	08/11/20
PMCM/Engineer:	Saudi Arabian Parsons Ltd	Project/Zone No:	L01
		Contractor:	MBL
Discipline:			
☐ Survey ☒ Civil ☐ Architectural ☐ Structural ☐ Landscape ☐ Electrical ☐ Low current ☐ Mechanical ☐ HSE ☐ Testing/Commissioning ☐ Others (.....) Sub-discipline (.....)			
Work Details:			
Description:			
Request for inspection of anti-termite at area 2 zone 3 L01 as per attached drawing.			
Approved Method Statement:		Section:	NN
Approved Inspection and Test Plan:		Clause:	NN
Clearance Form Enclosed:		☐ Yes / Disciplines (.....) ☐ No	
Approved Drawing No.:	L01-MBLC-D04-STR-DWG-10B250-F1-LAY100	Attached Approved Drawings	☒
Location / Level / Room Name & No. / Grid Line:	ZONE 3	Attached Location Sketch	☒
		Attached Check Sheet	☐
		Attached Previous Revision (if any)	☐
The above work has been completed according to Project requirements and specifications and is ready for inspection			
MBL Construction's Representative:			
Name:	Tarek Amer	Signature:	
Date:	07/11/20		
MBL QA/QC's Representative:			
Name:	Khaled Naim sabha	Signature:	
Date:	07/11/20		
Supervision Consultant's Review:			
Checks:			
Site Test Results (if any): ☐ Passed ☐ Failed ☐ Not Applicable			
Status of Inspection: ☒ Status A (Approved) ☐ Status B (Approved with Comments) ☐ Status C (Objection)			
Comments:			
+ Works was done for the above-mentioned & checked by the CSR team			
Reviewer Name:		Signature:	
Date:	20		
PMCM's Review:			
Endorsement: ☒ Yes ☐ No			
Comments:			



Verifying WIR and updating in model as per site conditions

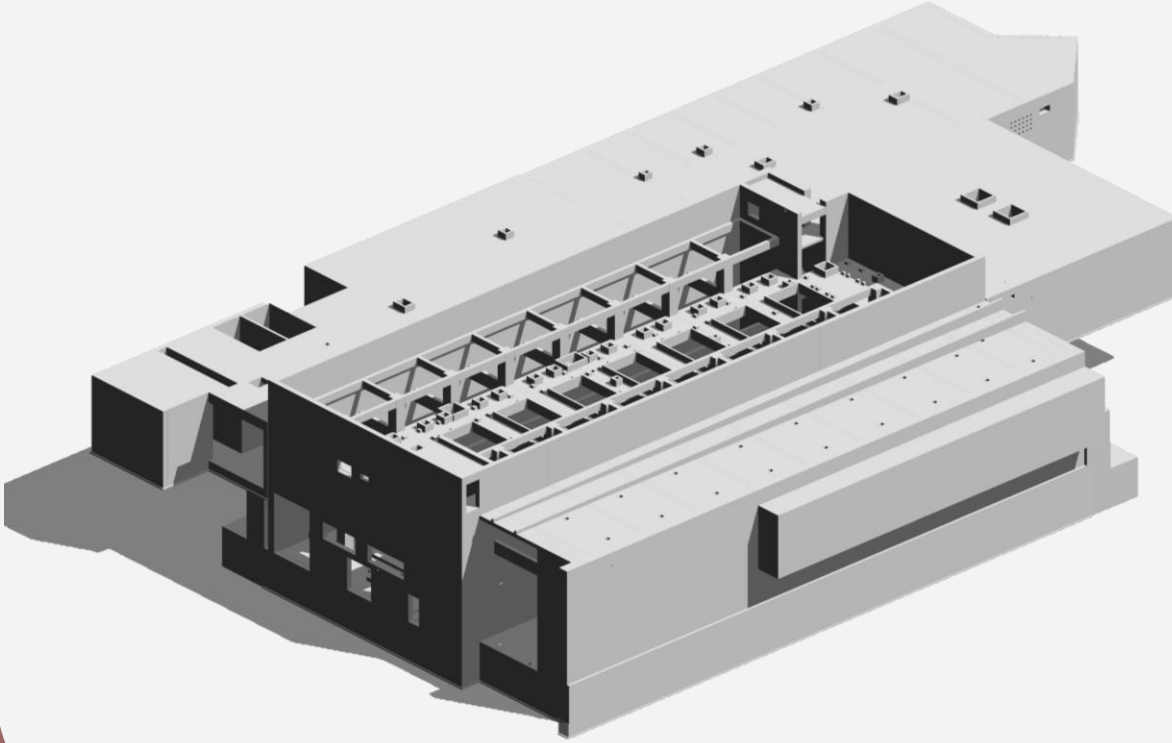


Exhibition

Highlighted projects with smart detailing and clean coordination deliverables

5

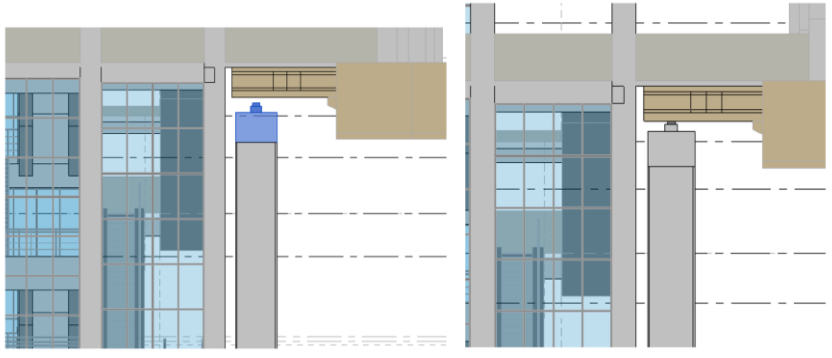
As Close As It Gets



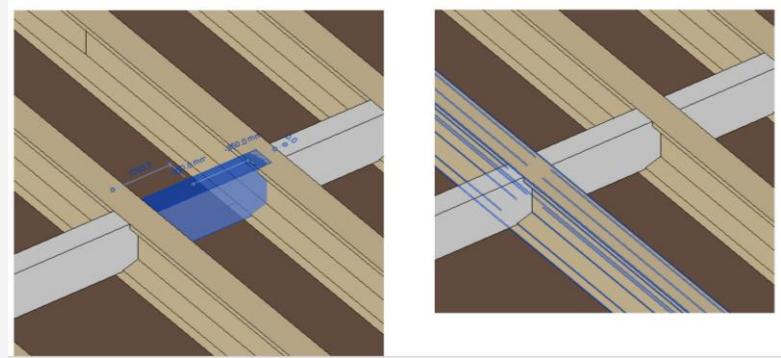
My role involves model development, clash detection, quantity take-offs, and ensuring compliance with BIM standards (LOD, COBie, IFC). And closely involved in coordination and review meetings, gaining practical insights into BIM workflows, collaboration, and project learning.

Documenting for clash detection and coordination.

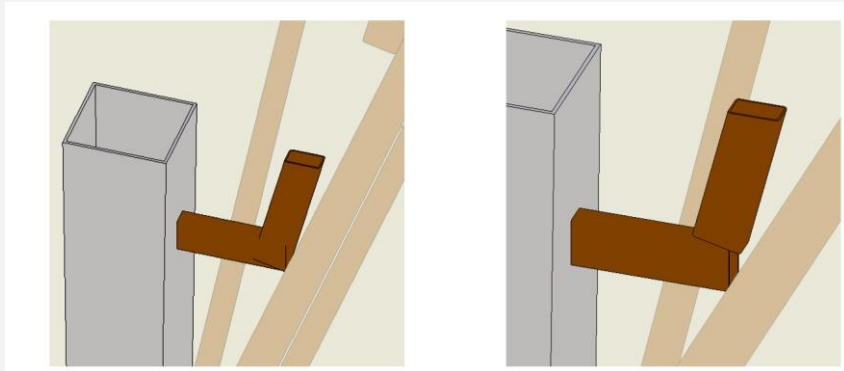
Support height change based on the other side



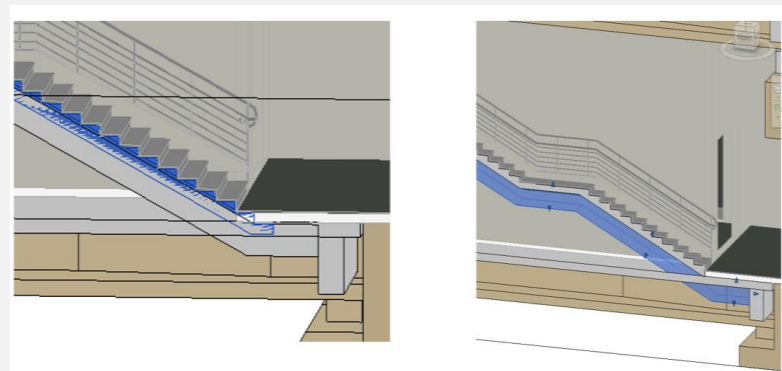
All diaphragm overlaps rectified

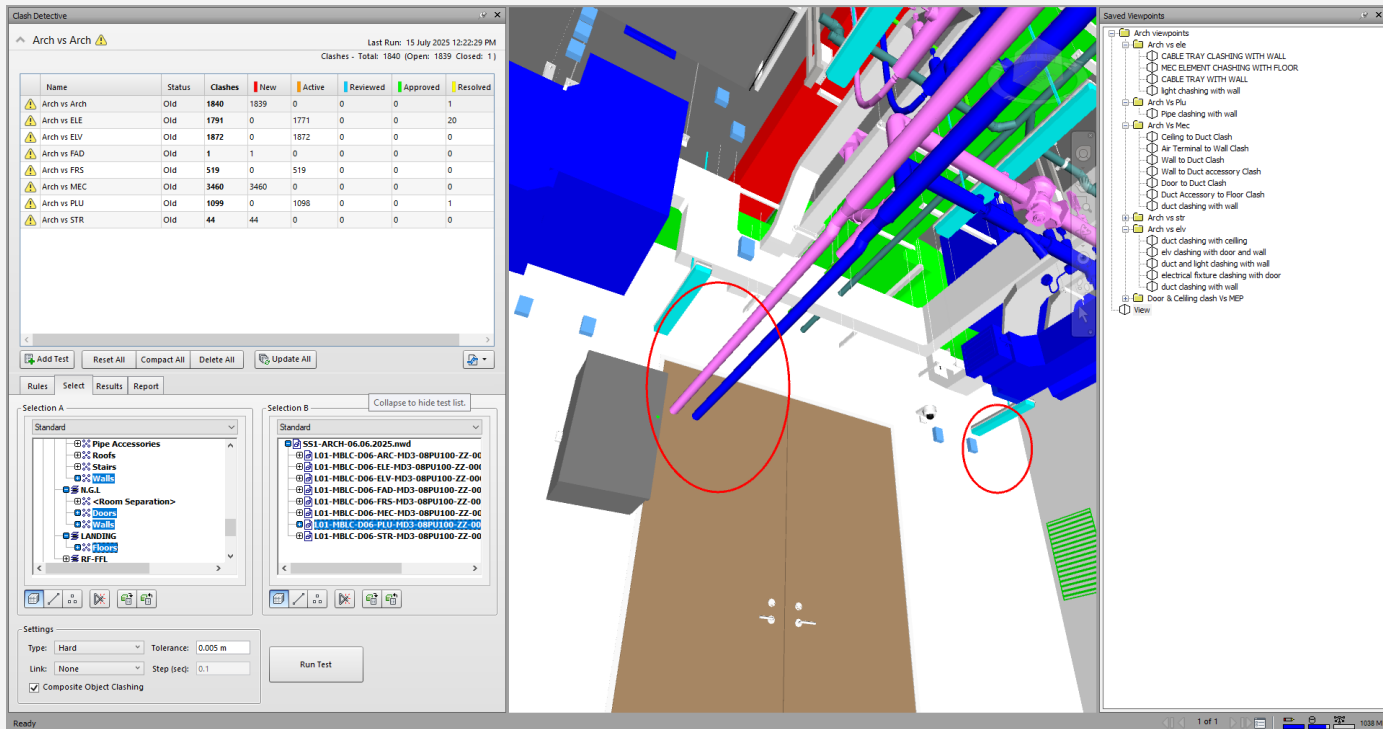


Interference changed

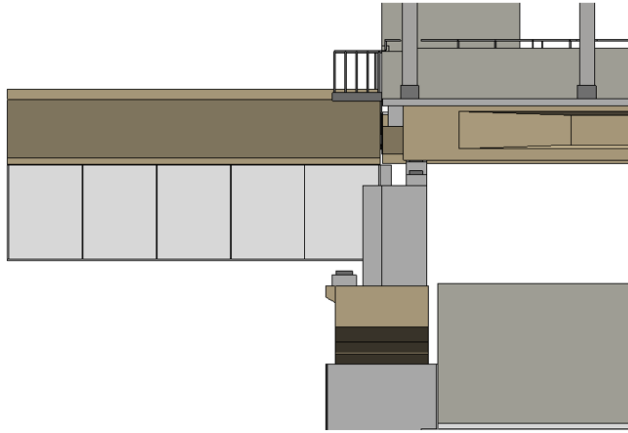


Stringer beam aligned to staircase



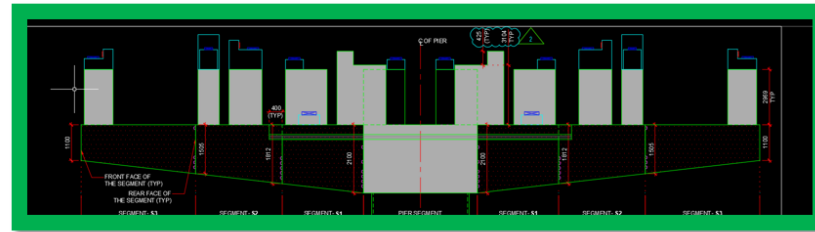


While working in Navis-works, whenever coordination doubts or clashes arose, I utilized saved viewpoints to document and communicate issues clearly with the team. This approach ensured that design conflicts were easy to identify, review, and resolve efficiently.



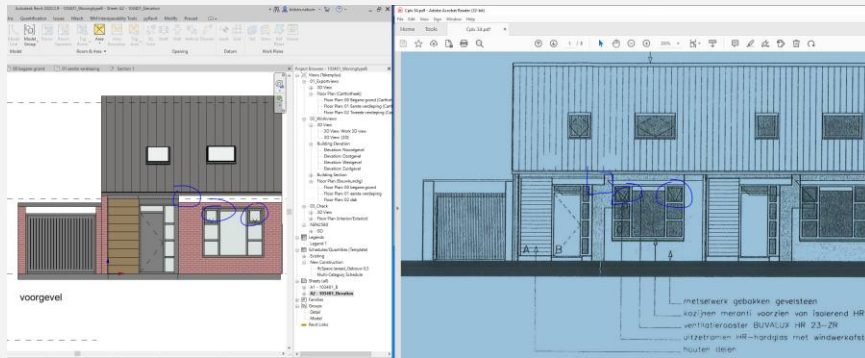
The girders are not resting on pedestal please provide the latest file .

Please refer the below sketch for the height of the stub columns. Even after correcting the heights then move the pier arm upward to make the girder to sit on the bearing.

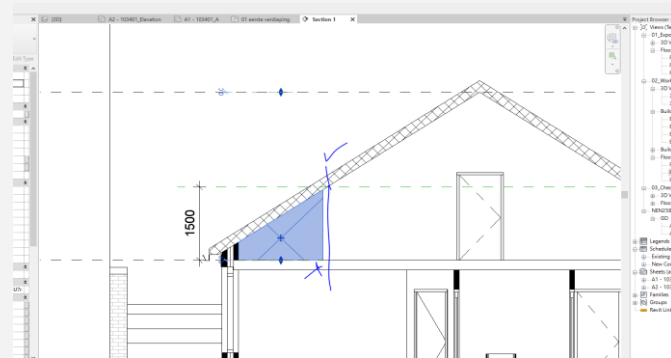


While coordination meetings are essential, conducting them every single time is not always feasible. Therefore, structured documentation serves as an efficient alternative for smooth communication and project alignment.

Client Requirement: QC and Precision at the Core of BIM Execution



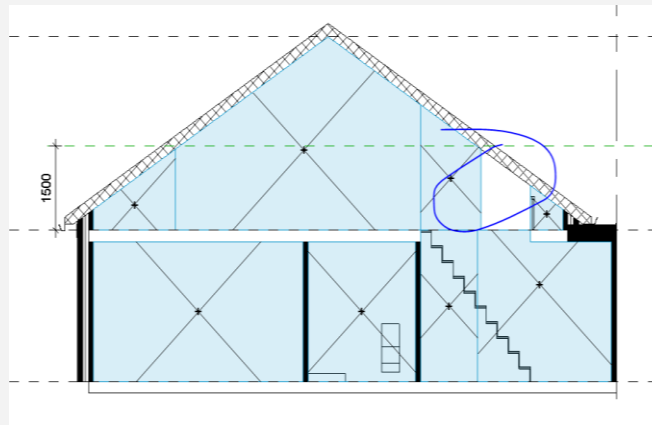
- Roof are not as per source
- Swings of window are not as per source



Room separation line not exactly on 1500mm cut plan

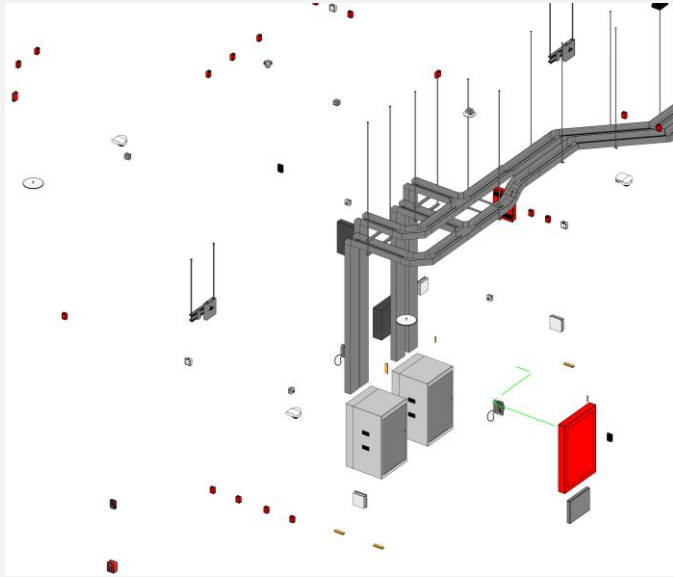
[illegible]

Wrong code for this exterior walls



Room Missing

As per client requirement, the project demanded rigorous QC checks where every parameter in Revit schedules had to be validated with precision.



- 01_FIRE ALARM SYSTEM
- 02_TETRA SYSTEM
- 03_TECHNOLOGY SYSTEM
- 04_GSM_TETRA
- 05_CCTV COVERAGE SYSTEM

ELV Support (Cross-Disciplinary Experience)

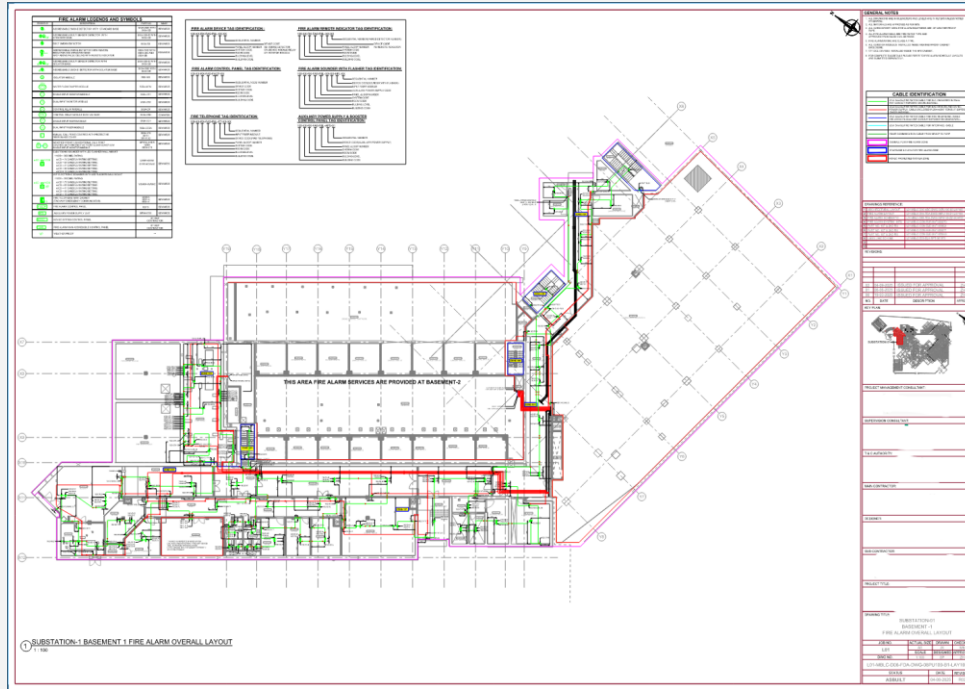
Stepped in to support the MEP team during tight deadlines on ELV services (CCTV, Fire Alarm, Wi-Fi, Containment, VCS).

Quickly adapted to new systems, overcoming initial challenges and completing the assigned scope.

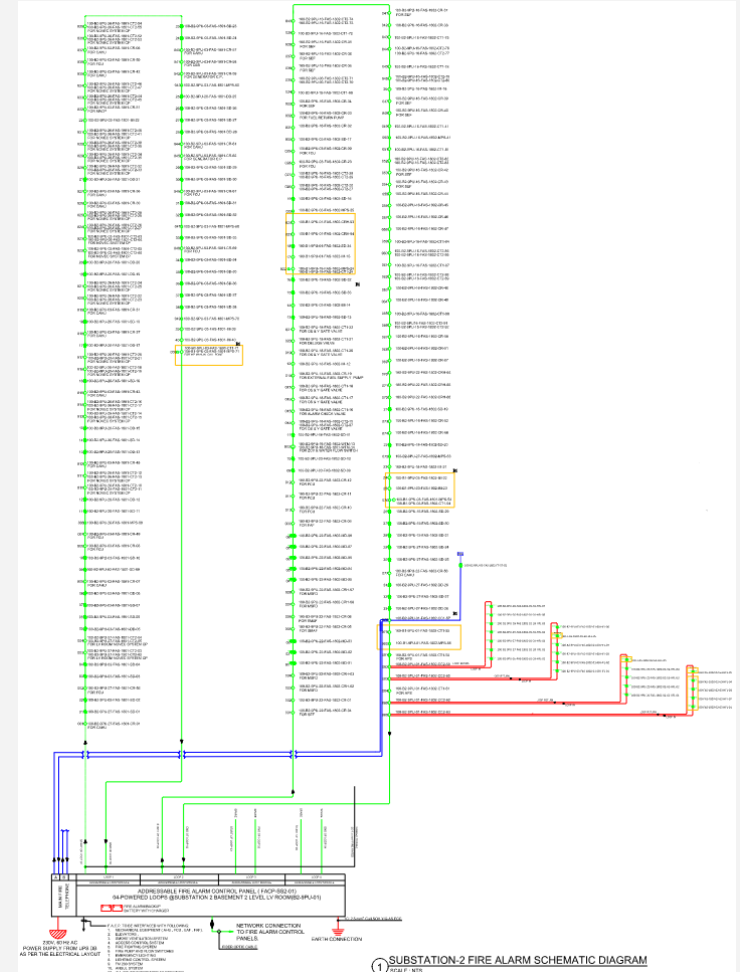
Enhanced technical versatility, problem-solving, and resilience under real project conditions.

Contributed to ELV services (CCTV, Fire Alarm, Wi-Fi, Technology-Containment, VCS,GSM and Tetra)

Sample of work on ELV



Developed schematic diagrams and based on Work Inspection Reports (WIR) updated the model to ensure design clarity and compliance.

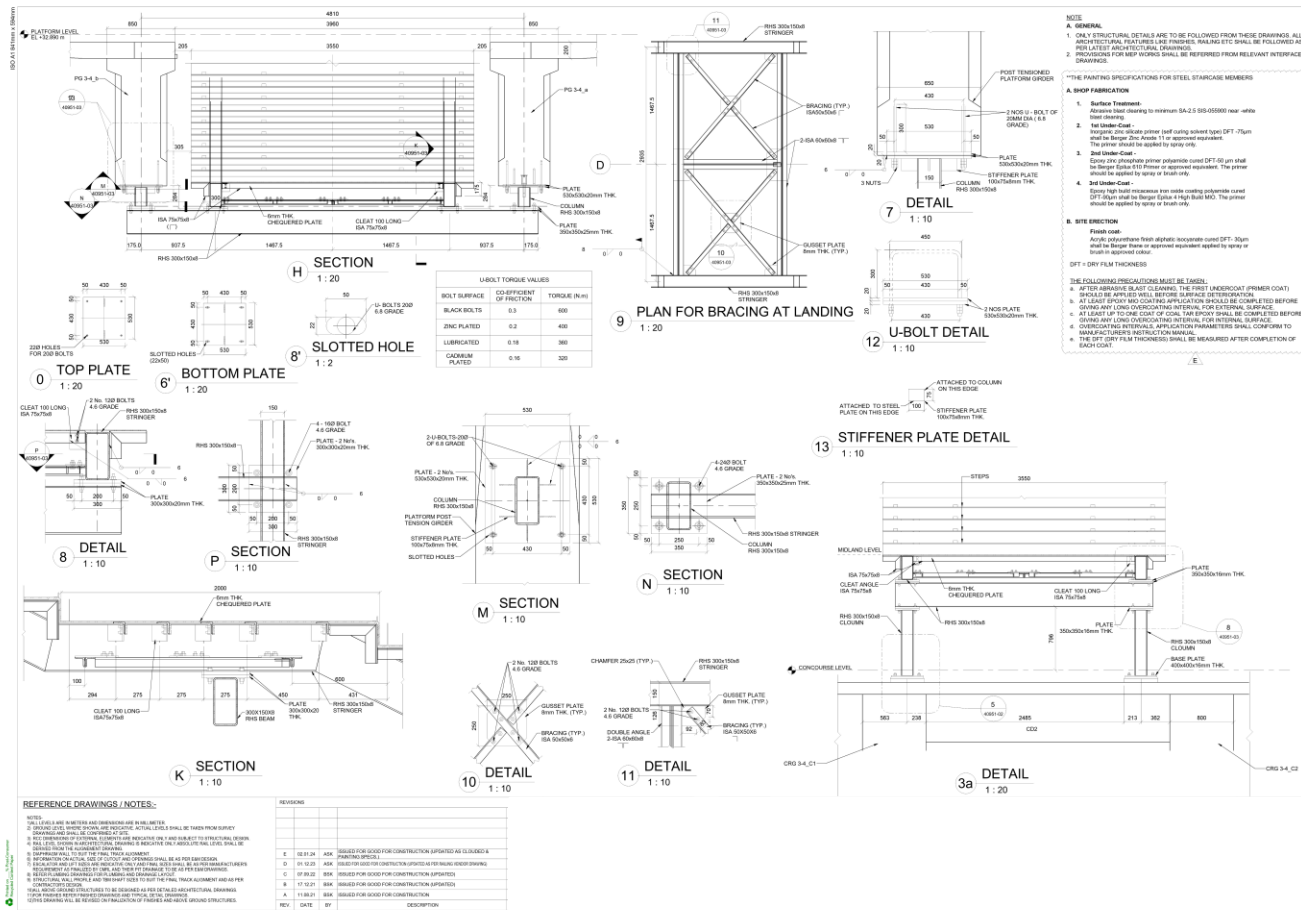




Documentation

Documentation as a Tool for Project Clarity

5.1



Schedule (BOQ)

LEGEND

[illegible]

Fire Alarm Device Schedule

SUBSTATION 3 FIRE ALARM DEVICE SCHEDULE (FACP-553-14/LOOP-1) GROUND FLOOR											
S. NO.	DEVICE TAG 1		DEVICE TAG 2		DEVICE TYPE	ROOM NAME	ROOM NUMBER	INTERFACES TO	SVL	JOIN	
1	100-00-12PU04-FAS-1401-CC1-01				1	SINGLE INPUT MONITOR MODULE	GROUND FLOOR	CONTROL ROOM	00-12PU-04	NOT APPLICABLE	100
2	100-00-12PU04-FAS-1401-CT2-02				2	SINGLE INPUT MONITOR MODULE	GROUND FLOOR	CONTROL ROOM	00-12PU-04	NOT APPLICABLE	100
3	100-00-12PU04-FAS-1401-CC2-03				3	DUAL INPUT RISER MODULE	GROUND FLOOR	CONTROL ROOM	00-12PU-04	NOT APPLICABLE	100
4	100-00-12PU04-FAS-1401-M10-01				1	ADDRESSABLE MULTI SENSOR DETECTOR	GROUND FLOOR	FOAM EQUIPMENT	00-12PU-01	NOT APPLICABLE	100
5	100-00-12PU02-FAS-1401-CT1-07				7	SINGLE INPUT MONITOR MODULE	GROUND FLOOR	FOAM EQUIPMENT	00-12PU-02	FOR OS & GATE VALVE	100
6	100-00-12PU02-FAS-1401-CT8-08				8	SINGLE INPUT MONITOR MODULE	GROUND FLOOR	FOAM EQUIPMENT	00-12PU-02	FOR OS & GATE VALVE	100
7	100-00-12PU02-FAS-1401-CT1-09				9	SINGLE INPUT MONITOR MODULE	GROUND FLOOR	FOAM EQUIPMENT	00-12PU-02	FOR OS & GATE VALVE	100
8	100-00-12PU05-FAS-1401-CT2-10	100-00-12PU05-FAS-1401-CT2-11			10	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-03	00-12PU-05	FOR NOVEC	100
9	100-00-12PU05-FAS-1401-CT2-12	100-00-12PU05-FAS-1401-CT2-13			12	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-05	FOR NOVEC	100
10	100-00-12PU06-FAS-1401-CT3-52	100-00-12PU06-FAS-1401-CT3-53			52	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
11	100-00-12PU06-FAS-1401-CT3-54	100-00-12PU06-FAS-1401-CT3-55			54	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
12	100-00-12PU06-FAS-1401-CT3-56	100-00-12PU06-FAS-1401-CT3-57			56	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
13	100-00-12PU06-FAS-1401-CT3-58	100-00-12PU06-FAS-1401-CT3-59			58	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
14	100-00-12PU06-FAS-1401-CT2-60	100-00-12PU06-FAS-1401-CT2-61			60	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
15	100-00-12PU06-FAS-1401-CT2-62	100-00-12PU06-FAS-1401-CT2-63			62	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR GENERATOR RUN IN TROUBLE MANUAL	100
16	100-00-12PU06-FAS-1401-CT1-18	100-00-12PU06-FAS-1401-MPPS-18			18	SINGLE INPUT MONITOR MODULE/MANUAL CALL POINT	GROUND FLOOR	TR-02	00-12PU-06	NOT APPLICABLE	100
17	100-00-12PU06-FAS-1401-IM-02				2	ISOLATOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	NOT APPLICABLE	100
18	100-00-12PU06-FAS-1401-CT2-15	100-00-12PU06-FAS-1401-CT2-14			15	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	NOT APPLICABLE	100
19	100-00-12PU06-FAS-1401-CT2-16	100-00-12PU06-FAS-1401-CT2-17			16	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR NOVEC	100
20	100-00-12PU06-FAS-1401-CT2-19	100-00-12PU06-FAS-1401-CT2-20			19	DUAL INPUT MONITOR MODULE	GROUND FLOOR	TR-02	00-12PU-06	FOR NOVEC	100

FIRE ALARM LEGENDS AND SYMBOLS			
SYMBOLS	DESCRIPTION	PART NO	MAKE
	DUCT SMOKE DETECTOR	SIGA-SD	EDWARDS
	ISOLATOR MODULE	SIGI-IM2	EDWARDS
	ADDRESSABLE MULTI SENSOR DETECTOR WITH STANDARD BASE	SIGA-PHD WITH SIGA-SB	EDWARDS
	SINGLE INPUT MONITOR MODULE	SIGA-CT1	EDWARDS
	DUAL INPUT MONITOR MODULE	SIGA-CT2	EDWARDS
	CONTROL RELAY MODULE	SIGA-CR	EDWARDS
	SINGLE INPUT RISER MODULE	SIGA-CC1	EDWARDS
	DUAL INPUT RISER MODULE	SIGA-CC2A	EDWARDS
	WIP MANUAL CALL POINT COVERED WITH PROTECTIVE FIBER GLASS COVER	MPSR2-SHTW-GE	EDWARDS
	FIRE TELEPHONE WITH CABINET (TWO WAY EMERGENCY COMMUNICATOR)	6831-4 6831-2 6831-4	EDWARDS
	FIRE ALARM CONTROL PANEL	EST3	EDWARDS
	AUXILIARY POWER SUPPLY UNIT	BPS6A/230	EDWARDS
	FIRE PROOF CONVENTIONAL CALL POINT COVERED WITH PROTECTIVE FIBRE GLASS COVER C/W SINGLE INPUT MONITOR MODULE	MPSR2-SHTW-GE WITH SIGA-CT1/ SIGA-CT1HT	EDWARDS
	MANUAL CALL POINT COVERED WITH PROTECTIVE FIBER GLASS COVER	SIGA-278	EDWARDS
	OUTDOOR ELECTRONIC SOUNDER WITH FLASHER WALL / CEILING MOUNT		
WP xxCD = 110DB 	110DB = DECIBEL RATING. xxCD = 65 CANDELA RATING/SETTING. xxCD = 60 CANDELA RATING/SETTING. xxCD = 50 CANDELA RATING/SETTING. xxCD = 41 CANDELA RATING/SETTING.	WG4RF-HV/RHC	EDWARDS

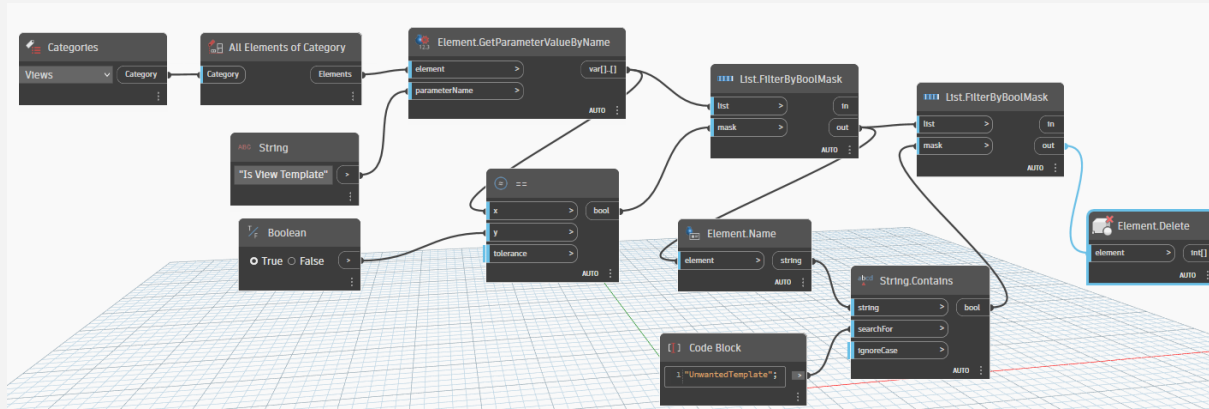


R&D-RESEARCH DEVELOPMENT

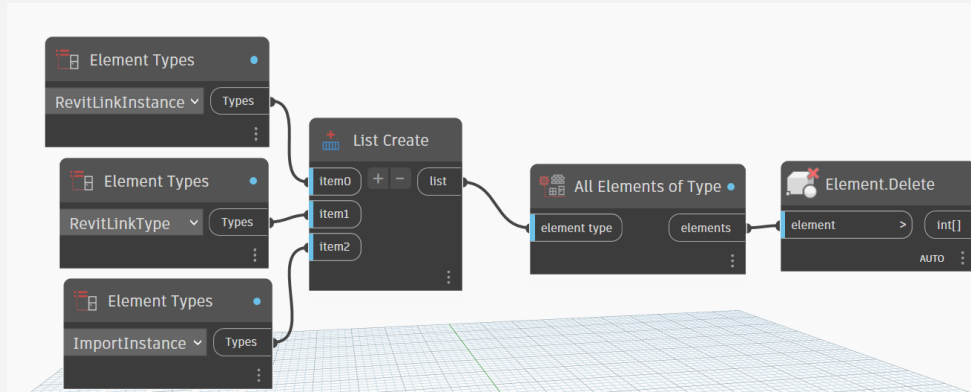
Advancing BIM Methodologies through Continuous Learning and
Application

6

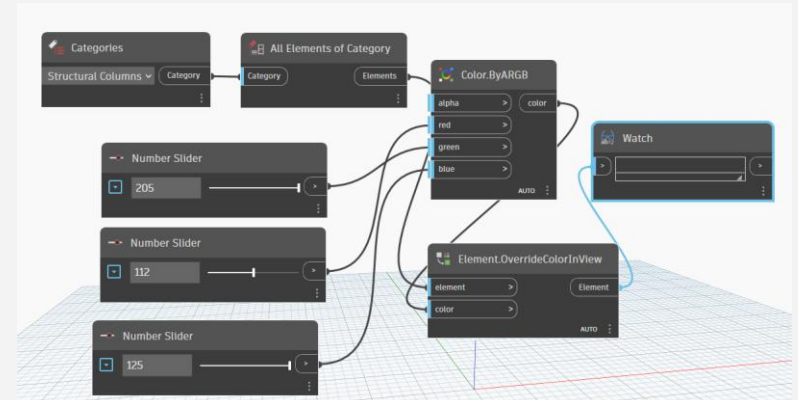
Script for Deleting Unused Templates



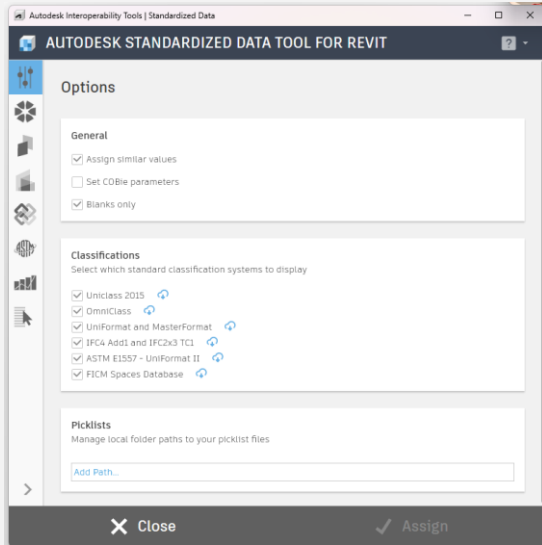
Explored AI-assisted scripting by collaborating with Chat-GPT to develop Dynamo scripts, improving automation and problem-solving skills in BIM environments.



Script for Purge Links



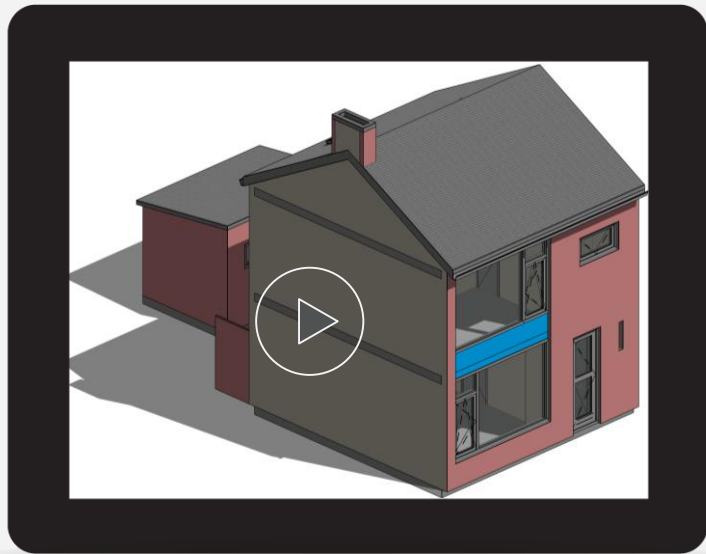
Script for Element color code



Gained foundational knowledge of Autodesk Interoperability Tools — COBie Extension, Classification Manager, and Standardized Data Tool — to support structured data exchange in BIM workflows.



Watch the full Synchro 4d Model Timeline Walkthrough



https://drive.google.com/file/d/1-ryV_fmDWbfih6AENzKWX45Nv0rjiLya/view?usp=sharing

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